

802.11 Wireless LAN Architecture

This diagram shows how a **Wi-Fi network (802.11 Wireless LAN)** works in a simple way.

What's happening:

- Each blue circle is called a **Basic Service Set (BSS)**
→ It's a small wireless network area.
- Inside each BSS:
 - There is an **Access Point (AP)** (the Wi-Fi router device)
 - Multiple **wireless devices** (laptops, phones) connect to that AP

Flow of communication:

1. Devices (like laptops) **connect wirelessly to the Access Point (AP)**
- ★ 2. The **AP acts as a base station** and collects all data from devices
3. The AP is connected (via cable) to a **switch or router**
4. The router connects everything to the **Internet**

Key idea:

- **BSS 1 and BSS 2 are two separate Wi-Fi zones**
- Both are connected to the same network (through the router)
- So, devices in different BSS can still communicate or access the internet

In short:

Devices → AP (wireless) → Router (wired) → Internet

Multiple APs = larger coverage area

IEEE 802.11 Infrastructure WLAN

- 1) 6-byte MAC address is stored in the firmware of the station's adapter.
- 2) Each AP has also a MAC address for its wireless interface.
- 3) APs have the wired connection with router

1. 6-byte MAC address in station (device)

- Every wireless device (laptop, phone) has a **unique MAC address**
- It is **6 bytes (48 bits)** long
- Stored in the device's **network adapter firmware**
- This address is used to **identify the device on the network**

Think of it like a **permanent ID** for your device

2. Access Point (AP) also has a MAC address

- Each **Access Point (AP)** has its own **MAC address**
- This MAC is used by devices to:
 - Identify which AP they are connecting to
 - Send/receive data correctly

So, both:

- Devices (stations) have MAC
- APs also have MAC

3. AP is connected to router using a wired link

- The AP is not standalone
- It connects to a **router or switch using a cable (Ethernet)**

Overall flow in Infrastructure WLAN:

- Device → (wireless) → AP
- AP → (wired) → Router → Internet

Key idea:

- **Wireless part** = Device ↔ AP
- **Wired backbone** = AP ↔ Router

In one line:

Devices use MAC addresses to communicate with APs, and APs forward data to the router through a wired connection.

IEEE 802.11 Ad-hoc networks

◇ What is an Ad-hoc Network?

An Ad-hoc network is a wireless network without an Access Point (AP).

- ✗ No central control
- ✗ No router / internet required
- ☒ Devices connect directly to each other

Also called a peer-to-peer network.

✈ ➤ A network with no central control and with no connections to the outside world.

◇ How communication happens:

- Each device acts like both:
 - a sender
 - and a receiver/router
- Data is sent directly from one device to another

Flow:

Device A ↔ Device B ↔ Device C (no AP in between)

◇ Real-life Examples:

1. Military / Disaster Communication
 - Soldiers' radios form a MANET (Mobile Ad-hoc Network)
 - Useful when no infrastructure exists
2. Group of laptops (no Wi-Fi available)
 - In places like:
 - conference rooms
 - trains
 - cars
 - Devices connect directly to share files
3. IoT sensors in remote areas
 - Sensors communicate with nearby sensors
 - Data hops from one node to another

◇ Difference from Infrastructure WLAN:

Feature	Infrastructure Mode	Ad-hoc Mode
Access Point	Required	Not used
Control	Centralized	Distributed
Internet Access	Usually yes	Usually no
Communication Via AP		Direct

◇ In one line:

Ad-hoc = devices talk directly without any AP or central network

Channels and Association in WLAN

Channels in WLAN

- In Wi-Fi (IEEE 802.11), the **Access Point (AP)** is configured by the network admin with:
 - **SSID** → Wi-Fi network name
 - **Channel number** → frequency used for communication
- The **2.4 GHz band (2.4–2.4835 GHz)** is divided into **11 channels**.
- These channels overlap, which can cause interference.
- To avoid interference, we use **non-overlapping channels**:
 - Best set: **1, 6, 11** ✓
(because they are far enough apart)

Association in WLAN

- **Association** = when a wireless device (phone, laptop) connects to an AP.
- ★ It creates a **virtual link (like a wireless cable)** between device and AP.
- After association, the device can send/receive data.

In one line:

- **Channels** → decide *which frequency Wi-Fi uses*
 - **Association** → *process of connecting device to Wi-Fi*
-

WiFi Jungle

- A WiFi jungle is a place where your device detects multiple Access Points (APs) at the same time.
- These APs may:
 - Belong to different networks (subnets)

- Use different or even same channels



- Example: apartment buildings, universities, malls.

How a Device Chooses an AP

How does your wireless device associate with a particular AP?

1. Beacon Frames

- In IEEE 802.11, every AP periodically sends beacon frames.
- Beacon contains:
 - SSID (network name)
 - MAC address
 - Other info (capabilities, timing)

2. Passive Scanning (default)

The process of scanning channels and listening for beacon frames is known as passive scanning

- Device scans all 11 channels (2.4 GHz)
- It listens for beacon frames from nearby APs
- If multiple APs are found:
 - Device usually connects to the strongest signal (highest RSSI)

3. Active Scanning

- Device broadcasts a probe request
- Nearby APs reply with probe responses
- Faster than passive scanning

Key Idea (1 line)

- WiFi jungle = many APs around
 - Device connects to AP with strongest signal using scanning (passive/active)
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Passive scanning

You see:

- AP1 and AP2 are broadcasting **beacons**
- Laptop H1 is just **listening**
- H1 receives signals from both APs
- It chooses one AP (based on strongest signal)
- Then:

1. Association Request → AP
2. Association Response ← AP

Flow is:

AP → H1 (beacons) → H1 chooses → connects

a. Passive scanning

1. Beacon frames sent from APs

2. Association Request frame sent:
H1 to selected AP

3. Association Response frame sent:
Selected AP to H1

strongest signal strength

Active scanning

You see:

- H1 sends **Probe Request** to both AP1 and AP2
- Both APs reply with **Probe Response**
- H1 compares responses
- Then it selects best AP
- Finally association happens:
 - Request → AP
 - Response ← AP

Flow is:

H1 → APs (probe) → APs reply → H1 chooses → connects

Key Difference (very important)

Feature	Passive Scanning	Active Scanning
Who starts?	AP	Device
Method	Listen to beacons	Send probe request
Power usage	Low	Higher
Speed	Slower	Faster
Use case	Normal background scanning	Quick network discovery

Simple way to remember

- **Passive** = “I wait and listen”
 - **Active** = “I shout and ask who is here?”
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802.11 MAC Protocol (From other slide)

CSMA

1) CSMA (Carrier Sense Multiple Access)

- Devices (A, B, C, D) listen before sending.
- At time t_0 , two devices think channel is free → both transmit.
- Due to propagation delay, they don't hear each other immediately.
- Result: Collision happens.

Key Idea:

Even after sensing idle, collision can still occur.

2) CSMA/CD (Collision Detection)

- Same situation as Pic 1, but now:
- Devices detect collision while transmitting.
- Once detected:
 - Transmission is aborted immediately
 - Saves time and bandwidth

Key Idea:

- Detect collision early → stop transmission → retry later
-

Binary Exponential Backoff

- After collision, devices wait random time before retrying.

Rule:

- Choose K from $[0 \rightarrow 2^n - 1]$
 - n = number of collisions
- Wait time = $K \times 512$, bit times (in Ethernet)

Meaning:

- More collisions → larger waiting range
- Reduces chance of repeated collision

Example:

- 1st collision → $K \in \{0,1\}$
- 2nd collision → $K \in \{0,1,2,3\}$

Key Idea:

More collisions = more delay = less future collision

CSMA/CA (Collision Avoidance) – Used in Wi-Fi (In IEEE 802.11)

Steps:

1. Channel idle?
 - Wait DIFS → then transmit
2. Channel busy?
 - Choose random backoff time
 - Countdown starts only when channel is idle
 - Freezes when busy
3. Transmit frame
 - Then wait for ACK
4. If no ACK
 - Assume collision/error
 - Retry with larger backoff

1. If initially the station senses the channel idle, it transmits its frame after a short period of time known as the Distributed Inter-frame Space (DIFS).

2. Otherwise, the station chooses a random backoff value using binary exponential backoff and counts down this value after DIFS when the channel is sensed idle. While the channel is sensed busy, the counter value remains frozen.

3. When the counter reaches zero, the station transmits the entire frame and then waits for an acknowledgment.

4. If the station has another frame to send, it begins the CSMA/CA protocol at step 2. If the acknowledgment isn't received, the transmitting station re-enters the backoff phase in step 2, with the random value chosen from a larger interval

Final Comparison (1 line each)

- CSMA → Listen before send (but collisions possible)
- CSMA/CD → Detect & stop collision (Ethernet)

- CSMA/CA → Avoid collision using backoff + ACK (Wi-Fi)
- Backoff → Random delay to reduce collisions

Link Layer Acknowledgment Scheme

This diagram shows a Link Layer Acknowledgment Scheme (used in Wi-Fi/802.11 communication).

Briefly:

- Source (sender) waits for a short time called DIFS before sending data.
- Then it transmits the data frame to the destination.
- Destination (receiver) waits a shorter time called SIFS (smaller than DIFS).
- After that, it sends an ACK (acknowledgment) back to confirm data was received correctly.

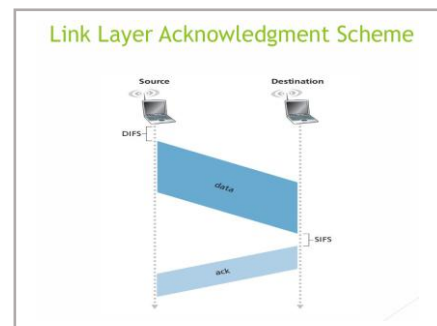
Key idea:

- DIFS → send data → SIFS → send ACK
- If ACK is not received, the sender assumes failure and retransmits.

This mechanism ensures reliable communication at the link layer.

- **SIFS is shorter** → used for quick responses like ACK
- **DIFS is longer** → used before sending new data

This priority helps ACKs go first, making communication reliable.



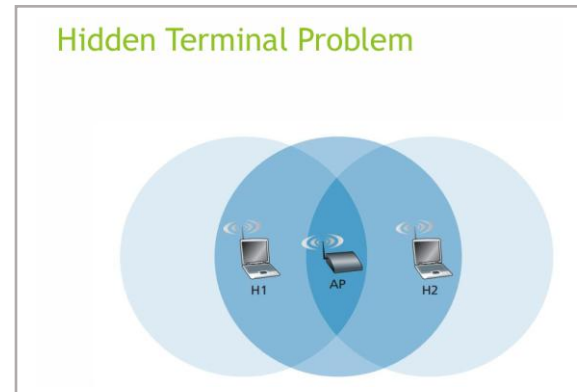
Hidden Terminal Problem in wireless networks.

What's happening:

- There are two devices: **H1** and **H2**, and a central **AP (Access Point)**.
- **H1 and H2 cannot hear each other** (they are outside each other's range).
- But both **can communicate with the AP**.

Problem:

- H1 thinks the channel is free → sends data
- H2 also thinks the channel is free → sends data
- Their signals **collide at the AP**



Solution (RTS/CTS):

1. **RTS (Request To Send)**
 - H1 sends RTS to AP before sending data
2. **CTS (Clear To Send)**
 - AP broadcasts CTS to all devices

Effect:

- H2 hears CTS → stays silent
- H1 sends data safely → **no collision**

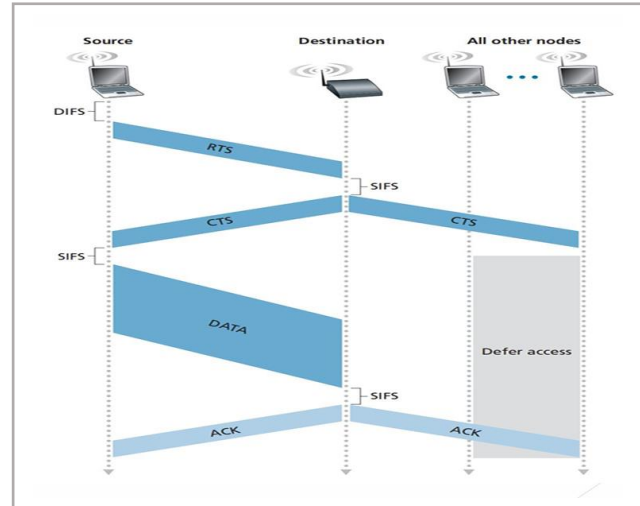
Simple idea:

Devices that can't hear each other → cause collision → RTS/CTS prevents it

Solution

1. Request to Send (RTS) control frame
2. Clear to Send (CTS) control frame

3. When a sender wants to send a DATA frame, it can first send an RTS frame to the AP, indicating the total time required to transmit the DATA frame and the acknowledgment (ACK) frame.
 4. When the AP receives the RTS frame, it responds by broadcasting a CTS frame.
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This diagram shows **RTS/CTS communication flow** in Wi-Fi to avoid collisions.

Step-by-step (brief):

1. **Source waits DIFS**
→ then sends **RTS (Request To Send)** to destination.
2. **Destination waits SIFS**
→ sends **CTS (Clear To Send)** back (also heard by other nodes).
3. **Other nodes hear CTS**
→ they **stop transmitting (defer access)** to avoid collision.
4. **Source sends DATA**
5. **Destination sends ACK** after SIFS
→ confirms successful delivery

Simple flow:

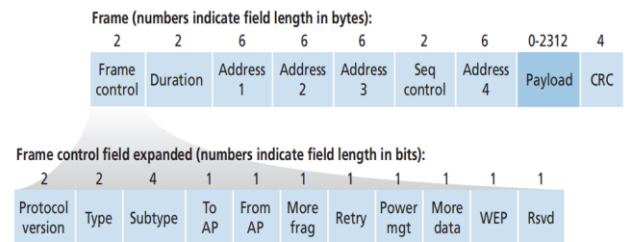
DIFS → RTS → SIFS → CTS → DATA → SIFS → ACK

Key idea:

RTS/CTS makes sure **only one device transmits**, while others stay silent → **no collision (hidden terminal solved)**

802.11 Frame

802.11 Frame



structure of a WiFi frame (802.11 frame) — basically how data is organized when sent over WiFi.

Main fields (top row)

Each block is a part of the frame:

- **Frame Control (2 bytes)** → tells what type of frame it is
- **Duration (2 bytes)** → how long channel will be used
- **Address 1 (6 bytes)** → Receiver MAC address
- **Address 2 (6 bytes)** → Sender MAC address
- **Address 3 (6 bytes)** → Router/AP or network-related address
- **Sequence Control (2 bytes)** → ordering of packets
- **Address 4 (6 bytes)** → extra address (special case)
- **Payload (0–2312 bytes)** → actual data
- **CRC (4 bytes)** → error checking

Frame Control (bottom part in pic)

This field is further divided into small flags:

- **Type/Subtype** → Data, Control, or Management frame
- **To AP / From AP** → direction of data
- **Retry** → if packet is resent
- **More data** → more packets coming
- **Power management** → device sleep/awake
- **WEP** → encryption used

Basically, this part tells **how to interpret the frame**

Why 4 Address Fields?

This is the main concept in your notes:

Normally (wired Ethernet):

- Only **2 addresses** → Sender & Receiver

But in WiFi:

There can be more devices involved, so we need up to **4 addresses**

Meaning of each address

- **Address 1 → Receiver**
 - Who should receive this frame
- **Address 2 → Transmitter**
 - Who is sending it right now
- **Address 3 → Network / Router (AP)**
 - Usually the router interface (gateway)
- **Address 4 → Used in special cases**
 - When APs forward frames **between each other** (like bridging / ad hoc)

1. The **fourth address** field is used when APs forward frames to each other in ad hoc mode.
2. The **second address** field holds the MAC address of transmitting node.
3. The **first address** field holds the MAC address of receiving node.
4. BSS is part of a subnet connected to any router interface.
5. **Address 3** contains the MAC address of this router interface

Simple Example

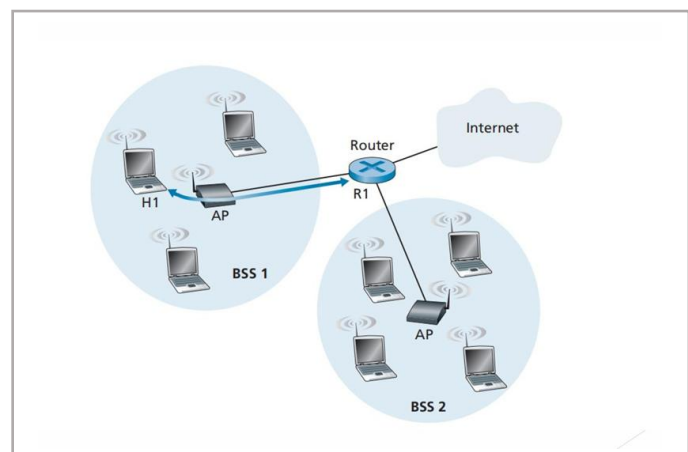
Think of delivery:

- **Address 1** → Final receiver (you)
- **Address 2** → Delivery guy (sender device)
- **Address 3** → Warehouse/router
- **Address 4** → Another warehouse (if forwarding between APs)

Key Idea

WiFi is more complex than wired networks because:

- Data may go **Device → AP → Router → AP → Device**
- So extra addresses are needed to track **who sent, who receives, and through whom**



What's going on in your picture

- You have **two WiFi cells (BSS1 and BSS2)**, each with an **AP (Access Point)**.
- Both APs are connected to a **router (R1)**, which connects to the **Internet**.

- **H1** is a wireless device inside **BSS1**.

Flow shown: **H1 → AP → Router (R1) → Internet**

What your text is explaining (H1 receiving data)

1. Router **R1** knows **H1's IP**, but not its MAC
2. So R1 uses **ARP** → finds H1's MAC address
3. R1 creates an **Ethernet (802.3) frame**
4. Frame goes to AP
5. AP **converts Ethernet frame → WiFi (802.11 frame)**
6. AP sends it wirelessly to H1

Important:

- **Address 3 = MAC of Router (R1)**
 - So H1 knows: *"this data came from router"*
-

What happens when H1 responds to R1?

Step-by-step (H1 → R1)

1. H1 creates data (e.g., request to internet)

- It already knows router IP (default gateway)
- It uses **ARP (if needed)** to get router MAC

2. H1 sends WiFi (802.11 frame) to AP

Frame addresses:

- **Address 1 (Receiver)** → AP
- **Address 2 (Transmitter)** → H1
- **Address 3 (Destination)** → Router (R1)

3. AP receives frame

- Converts **802.11** → **Ethernet (802.3)**
- Forwards it to router R1

4. Router (R1) receives it

- Processes the packet
 - Sends it to Internet (if needed)
-

Simple idea

- When **Router** → **H1**
→ AP inserts **router MAC** in **Address 3**
- When **H1** → **Router**
→ H1 puts **router MAC** in **Address 3**

Address 3 always helps identify the **other side of the network (router/gateway)**

One-line summary

- **Downlink (R1 → H1):** AP fills Address 3 with router MAC
 - **Uplink (H1 → R1):** H1 puts router MAC in Address 3
-